ASSIGNMENT 5 REPORT

Technical Analysis using R, Development Phase

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- **Course Code & Title:** BDA400 / Data Science Tools and Techniques
- **Assignment Title:** Assignment 5 - Technical Analysis using R, Development Phase
- **Repository Link:** [Insert your GitHub repository link here]

1. Overview of Implemented Technical Indicators

- **Simple Moving Average (SMA):** Computes the unweighted mean of data points over a specified period to smooth out short-term price fluctuations and identify general market trends.
- **Exponential Moving Average (EMA):** Similar to SMA but assigns greater weight to recent data points, making the indicator more responsive to latest price movements.
- **Moving Average Convergence Divergence (MACD):** Measures momentum and trend direction by calculating the difference between a short-term and long-term EMA, complemented by an EMA signal line and histogram.
- **Standard Deviation (stdev):** Quantifies data dispersion and volatility by calculating the square root of the variance of data points from the mean.
- **Linear Regression (linreg):** Models the linear relationship between variables to determine the slope, intercept, and predicted values for a given regression window.
- **Relative Strength Index (RSI):** A momentum oscillator measuring the speed and change of price movements to identify overbought (>70) or oversold (<30) conditions.
- **Stochastic RSI (StochRSI):** Applies the stochastic oscillator formula to RSI values to generate normalized %K and %D lines for sensitive overbought/oversold detection.

- **Crossover Function:** Detects bullish intersection events where one data series crosses above another, returning signal indicators ('Up', 'None').

- **Crossunder Function:** Detects bearish intersection events where one data series crosses below another, returning boolean validation signals.

2. Instructions on How to Run the Code

1. Open **RStudio** on your computer.

2. Set your working directory to the folder containing your .R script files using the command: setwd("path_to_your_folder").

3. Load dependent functions first if testing sequentially (e.g., ensure sma.R and ema.R are loaded before running dependent functions like macd.R or stoch_rsi.R).

4. Execute each script file individually in the R console or run the test examples provided at the bottom of each script to verify outputs.